# Do Now · retrieval

## Floating-point representation and normalisation

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

Write −2 in 8-bit two's complement.

### Q2

What is unsigned 00100000 after a logical right shift by one?

### Q3

Convert binary $0.1_2$ to decimal.

### Q4

Calculate $0.5 \times 2^3$.

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Why is 0.0110 not normalised? — For a positive two's-complement fractional mantissa, a normalised value begins with 01, not 00.

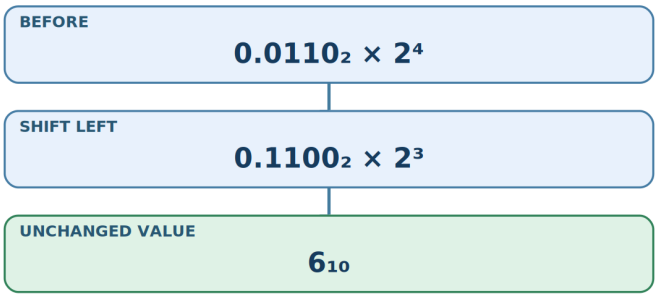
## Core idea 2

Shift and compensate — Shift 0.0110 left one place to 0.1100. Decrease the exponent from 4 to 3 to compensate.

## Core idea 3

Verify the value — $0.0110_2$ is 0.375; 0.375 × 16 = 6. $0.1100_2$ is 0.75; 0.75 × 8 = 6.

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.

# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

mantissa — The signed significant part of a floating-point number.

exponent — The power of two scaling the mantissa.

normalised — A two's-complement fractional mantissa with different first two bits.

precision — Number of significant bits represented.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

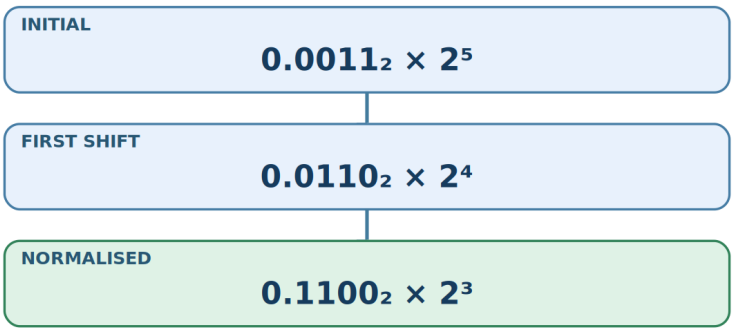
## Problem

Normalise positive $0.0011_2 \times 2^5$.

## Follow the worked method

1. First shift — 0.0011 becomes 0.0110; reduce exponent 5 → 4. It still begins 00.
2. Second shift — 0.0110 becomes 0.1100; reduce exponent 4 → 3. The first two bits are now 01.
3. Check — Both represent 6 in decimal, so normalisation preserved the value.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.

# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Predict the shift
Shift the mantissa left one position: 0.0101 → 0.1010.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Identify mantissa and exponent in $0.1100_2 \times 2^3$.

## Question 2

Normalise positive $0.0010_2 \times 2^5$.

Check that you have used the technical terms and explained your decisions.

# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

Why can normalisation improve precision without increasing the number of mantissa bits?

Check that you have used the technical terms and explained your decisions.

# Exit Ticket · reflect and retrieve

## Exit Ticket

Shifting $0.0110 \times 2^4$ left to $0.1100 \times 2^4$ preserves its value.

## Use key vocabulary from memory

1. Explain mantissa in your own words.

2. Explain exponent in your own words.

3. Explain normalised in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.